Rishit Roy

hellorishit.roy@gmail.com | 7678414702 | Greater Noida, India | github.com/rishit1711

July 22, 2026

Dear Hiring Team,

I'm writing to apply for the Software Engineering Apprenticeship at American Express. I'm a Computer Science (AI) undergraduate, and what drew me to this role is exactly what the description lays out — writing production code, working inside an agile team on API specs and code reviews, and building customer-facing systems that run at real scale. That's the kind of work I've been chasing on my own, outside of coursework.

Most of my hands-on experience is in Java and Spring Boot. I built an Airbnb-style backend system — hotel, room, inventory, booking, and payment modules — where the hard part wasn't the CRUD, it was getting concurrent bookings and pricing logic to behave correctly under transactional constraints. That project taught me more about REST API design and PostgreSQL than any single course did.

More recently, I've been building an AI Interview Copilot: it ingests PDFs, chunks the text, generates embeddings, stores them in a vector database, and runs a RAG pipeline to answer questions grounded in the document. I'm picking up Spring AI and vector databases along the way, mostly by breaking things and figuring out why. Given Amex's emphasis on proof-of-concepts, automation tooling, and data-driven decisions, that build-test-fix instinct feels like exactly what an apprenticeship needs.

I've also solved 300+ problems on LeetCode, mostly in Java, which keeps my data structures and algorithms sharp for the design and debugging work real engineering teams handle every day.

What draws me to Amex specifically is scale — a company processing this much data and this many transactions can't afford sloppy backend design, and that's the part of engineering I enjoy most: making systems correct and fast under real constraints. I'd welcome the chance to spend the next year learning how a team with Amex's track record does that, and contributing wherever I can.

I'd appreciate the opportunity to discuss this further. Thank you for considering my application.

Sincerely,
Rishit Roy